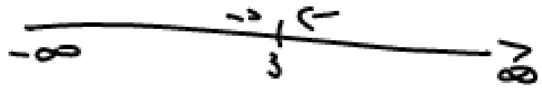
$f(x) = \dfrac{x - 2x^2 + 5}{3 - x}$, $D = x \in \mathbb{R} \setminus \{3\}$



$$\lim_{x\to\infty} \frac{x^2\left(1/x - 2 + 5/x^2\right)}{x\left(3/x - 1\right)} = x \cdot \infty$$ → diagonale Asymptote

$(-2x^2 + x + 5) : (-x + 3) = 2x + 5 - \dfrac{10}{-x+3}$

$-(-2x^2 + 6x)$

$\quad -5x + 5$

$\quad -(-5x + 15)$

$\qquad -10$

$\dfrac{10}{-x+3} \xrightarrow{\lim} 0$

→ $Asy(x) = 2x + 5$

$\dfrac{0}{0}$ → L'Hospital

$$\lim_{x\to 2} \frac{x^3 + 3x^2 - 6x - 8}{x^2 + x - 6} = \frac{0}{0}$$

$$\lim_{x\to 2} \frac{3x^2 + 6x - 6}{2x + 1} = \frac{12 + 12 - 6}{4 + 1} = \frac{18}{5}$$

$$\lim_{x \to 2} \frac{x-2}{2\sqrt{6-x}-4} = 0/0$$

L'Hospital

$$f(x) = a \cdot \heartsuit^n$$

$$f'(x) = a \cdot n \cdot \heartsuit^{n-1} \cdot \heartsuit'$$

$$\sqrt{6-x} = (6-x)^{1/2}$$

$$\Rightarrow 1/2 \cdot (6-x)^{1/2-1} \cdot (6-x)'$$

$$1/2 \cdot (6-x)^{-1/2} \cdot (-1)$$

$$-\frac{1}{2\sqrt{6-x}}$$

$$\lim_{x \to 2} \frac{1}{-\frac{\cancel{2}}{\cancel{2}\sqrt{6-x}}}$$

$$\Downarrow$$

$$\frac{1}{-\frac{1}{\sqrt{6-2}}} = -2$$

$$f(x) = \frac{3}{\sqrt[4]{2-x^2}} = 3(2-x^2)^{-1/4}$$

$$f'(x) = 3 \cdot (-1/4) \cdot (2-x^2)^{-1/4-1} \cdot (2-x^2)' = -\frac{3}{4}(2-x^2)^{-5/4}(-2x)$$

$$= 3/2\, x(2-x^2)^{-5/4} = \frac{3x}{2\sqrt[4]{(2-x^2)^5}}$$